

Debt Constraints in Emerging Economies: Evidence from Guatemala

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Abstract

I evaluate whether an institutional rule limiting debt monetization can credibly bind fiscal policy in a developing economy, using Guatemala's 1991 Monetary Board resolution and 1993 constitutional reform, which restricted central government borrowing from the central bank. Using the synthetic control method, I find that Guatemala's debt-to-GDP ratio diverged below its synthetic counterfactual after 1991, though the divergence is not statistically significant in most post-treatment years. The answer is more complex than a simple success or failure: the rule ended central-bank financing, but it did not produce a more conservative fiscal path. The government kept borrowing through other channels, and total public debt continued to rise. The decline in the debt-to-GDP ratio came from the denominator: inflation pushed nominal GDP well beyond the debt stock, so the improvement reflects an inflation-tax adjustment rather than fiscal discipline.

Keywords: fiscal rules, debt monetization, inflation tax, synthetic control, Guatemala

JEL Classification: H63, E58, E63, O23

1 Introduction

Why do some countries fall into debt traps while others maintain fiscal discipline? Over the past two centuries, government debt has evolved from a temporary emergency tool to a routine mechanism for financing policy ambitions (Buchanan and Wagner 1977). While this transformation has enabled governments to expand public investment, it has also introduced vulnerabilities, particularly for developing nations, where fiscal capacity and institutional credibility are weaker. As a counterbalance, legal or institutional mechanisms can limit governments' ability to accumulate debt excessively, including constitutional or statutory fiscal rules.¹

Among these fiscal rules are those that restrict debt monetization, the practice of financing deficits through central bank credit. While convenient in the short term, this strategy is widely associated with inflationary pressures and long-term macroeconomic instability (Andolfatto and Spewak 2018; Kwon et al. 2009). Limiting governments' ability to rely on central bank financing is thus a cornerstone of credible fiscal frameworks.

This paper asks whether institutional rules that limit debt monetization can credibly bind fiscal policy in a developing country. I evaluate the effectiveness of Guatemala's early 1990s fiscal rules that limited central government borrowing from the central bank. Using the Synthetic Control Method (SCM), I focus primarily on the 1991 Monetary Board resolution as the key policy shock, and treat the 1993 constitutional amendment as a reinforcement of the post-1991 shift rather than a separate policy shock. This approach allows me to construct a counterfactual of what Guatemala's debt trajectory would have been in the absence of the reforms. The findings show a divergence in debt trajectories following the 1991 reform, with Guatemala exhibiting lower debt-to-GDP ratios than the synthetic counterfactual. However, throughout the whole post-treatment period there is only one year with statistical significance (2000). The lowering of debt-to-GDP did not come from the numerator (debt), but the denominator (nominal GDP): the debt was inflated away, while the exchange rate stayed relatively stable. Therefore, the government was able to continue acquiring new debt, while lowering its debt-to-GDP ratios.

This paper contributes to the literature by showing that binding one financing margin, central-bank credit to the government, does not constrain the overall debt path: the government continued to accumulate debt through other channels, and the debt-to-GDP ratio fell

¹Many countries have introduced fiscal rules. In advanced economies, examples include Colorado's Taxpayer Bill of Rights (TABOR) in the United States, Switzerland's 2003 federal debt brake, Germany's 2009 Schuldenbremse, and the European Union's Stability and Growth Pact, which are often cited as successful in promoting fiscal discipline. Beyond high-income countries, several emerging economies have also implemented fiscal rules, including Panama (2006), Peru (2000 and 2003), Colombia (2011), and many CFA zone countries in Africa. In most of these cases, the rules have helped curb rising debt levels.

through the inflation tax rather than fiscal consolidation.

2 Historical Context

To understand this shift in policy towards better fiscal management, one must examine the preceding economic and political context. The late 1970s and 1980s were marked by debt crises, global interest rate shocks, and runaway inflation across Latin America. Central America faced these macroeconomic pressures while simultaneously undergoing political transitions and internal conflicts (Berntzen 1993; Devlin and Ffrench-Davis 1995).

Guatemala, throughout its history, experienced many major episodes of macroeconomic instability (Maul et al. n.d.). The latest occurred between 1986 and 1990: as the country transitioned to democracy, the government pursued expansionary fiscal and monetary policies, including routine debt monetization. Inflation surged, peaking near 60%, and the fixed exchange rate collapsed in 1989. These outcomes signaled the unsustainability of Guatemala’s macroeconomic framework.

In response, two major reforms were introduced to limit central bank financing of government debt.² In 1991, the Monetary Board (Junta Monetaria) adopted a rule requiring the government to repay the central bank more than it borrowed, effectively reducing net central bank lending (Banco de Guatemala 2024a). Two years later, a 1993 constitutional reform elevated this restriction to the highest legal hierarchy, prohibiting central bank credit to the government completely except in emergencies. Implemented in 1994, this reform institutionalized the earlier shift and made a reversal politically and legally costly. Since then, Guatemala has largely maintained fiscal discipline, with the only notable exception occurring in 2020 during the COVID-19 pandemic.

Figure 1 shows two measures of central-bank financing of the government: net credit to the public sector and the stock of bonds held by the Bank of Guatemala (Banco de Guatemala n.d.). Both peaked around 1991 and then fell sharply: net credit turned negative by 1994 as government deposits came to exceed central-bank credit, and the bond stock was settled by 1997. Taken together, the crises of the 1980s and the subsequent reforms created a natural experiment to study whether the rules implemented could credibly constrain fiscal policy. This sequence of events sets the stage for a research design that treats the 1991 resolution as a natural policy intervention and evaluates its impact on public debt using the Synthetic Control Method (SCM).

²A previous attempt in 1986, the Program for Social and Economic Reorganization (Programa de Reordenamiento Económico y Social, PRES), cautiously proposed reducing central bank financing, but it had little effect because it lacked clear targets.

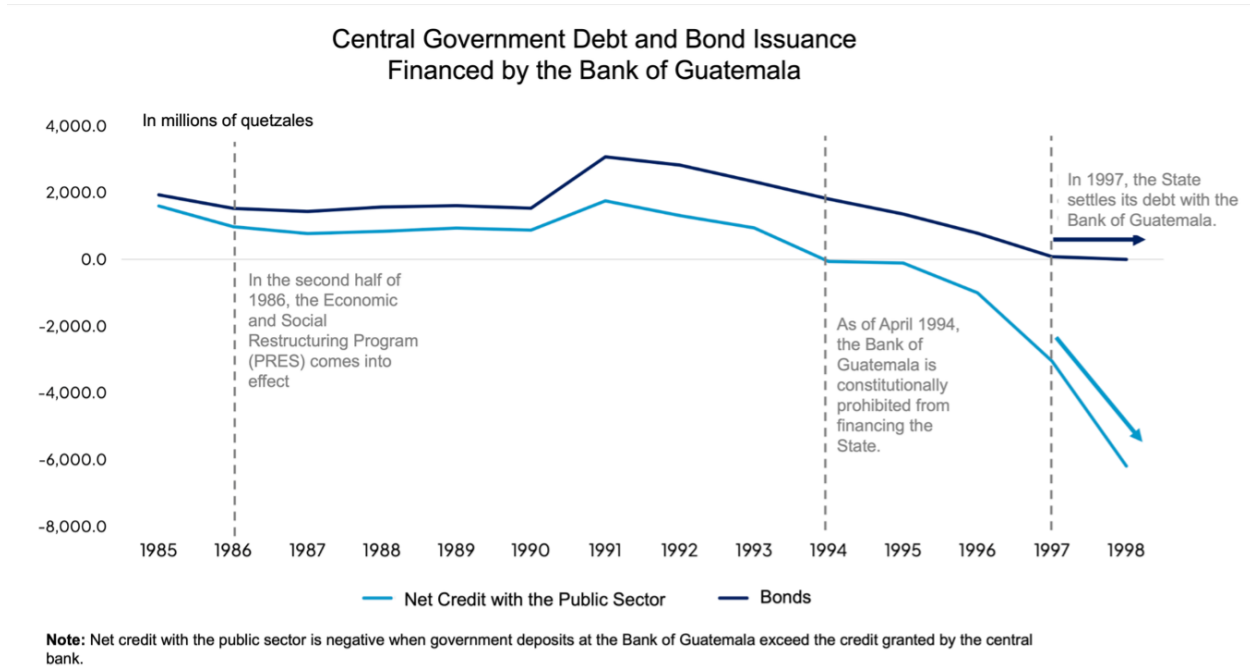


Figure 1: Central Government Debt and Bond Issuance Financed by the Bank of Guatemala (1985 to 1998)

Source: Banco de Guatemala (2024). Evolución de la política crediticia en Guatemala. Graph translated by the author.

3 Fiscal Rules, Debt Monetization, and Institutional Design

Governments have three main ways to finance their expenditures: taxation, borrowing, and money creation (debt monetization). Taxation has political limits, as voters resist higher taxes, while borrowing shifts costs to future generations but raises concerns about sustainability. When these options are exhausted, governments frequently resort to printing money, which can generate short-term relief but carries long-term risks of inflation and financial instability. The challenge is that political incentives often favor the path of debt, and even monetizing it: leaders face immediate pressure to spend, but the costs of financing are delayed and widely dispersed, creating what Buchanan and Wagner (1977) call a structural deficit bias.

Debt monetization is particularly dangerous in countries with weak institutions, where it often fuels inflation and undermines macroeconomic credibility. Andolfatto and Spewak (2018) argue that while monetization can be justified during crises, its normalization encour-

ages overborrowing and weakens central bank independence. Kwon et al. (2009) find that money-financed debt is strongly associated with inflation, especially in developing economies. Sumner (2021) distinguishes between crisis-driven monetization (manageable under credible institutions) and chronic monetization, which becomes entrenched and erodes trust. The key determinant is institutional credibility: weaker institutions face harsher consequences when relying on monetization.

Public Choice theory offers a framework for addressing this problem of deficits more broadly. Since governments are composed of self-interested actors with incentives to overspend and defer fiscal costs, there is a need for constraints that will lead policymakers' decisions towards long-term stability. Brennan and Buchanan (1980) advocated for constitutional constraints on government fiscal behavior. They proposed that by embedding fiscal discipline into the highest legal authority (a constitution), rather than relying on short-term political will, governments could be prevented from exploiting their power to tax and borrow. Wyplosz (2011) similarly argued that formal fiscal rules with legal or constitutional backing are more enforceable than informal norms, and more likely to succeed over time.

These theoretical insights emphasize the importance of legal hierarchy in making fiscal rules credible. Rules embedded in higher legal frameworks, such as constitutions, raise the political cost of reversal and signal stronger commitment to fiscal discipline (Brennan and Buchanan 1980; Wyplosz 2011). This perspective is particularly relevant for Guatemala, where a 1991 Monetary Board resolution first limited government borrowing from the central bank and a 1993 constitutional amendment subsequently prohibited such credit entirely. This context allows me to study whether raising a fiscal constraint to the constitutional level enhances its credibility and strengthens fiscal outcomes.

3.1 Fiscal Rules, Fiscal Outcomes, and Econometrics

A large body of research has analyzed fiscal rules and how these affect fiscal outcomes using a variety of econometric approaches, ranging from panel data to quasi-experimental methods. Surveys like Brändle and Elsener (2024) find that fiscal rules are generally associated with better fiscal outcomes, such as lower deficits, reduced debt ratios, less spending volatility, more accurate fiscal forecasts, and improved sovereign bond ratings. Most of these studies used panel data regressions, dynamic GMM estimations, and even Instrumental Variables (IV). This consensus has shaped the view that fiscal rules are desirable and an essential tool for achieving fiscal sustainability.

However, not all studies agree with this view. For example, when analyzing the impact of Colorado's Taxpayer Bill of Rights (TABOR), Eliason and Lutz (2018) find no statistically

significant impact on spending or taxes using the Synthetic Control Method. They argue that the rule only formalized previously-existing preferences. They mention the possibility that part of the positive evidence of the effect of fiscal rules might have happened because of endogeneity: countries with a more prudent fiscal approach might be more likely to adopt fiscal rules, so the results would be biased upward.

Since Eliason and Lutz’s study, a new literature has emerged studying fiscal rules using quasi-experimental methods, mainly the Synthetic Control Method (SCM). This econometric method, developed by Abadie and Gardeazabal (2003) in their study of the Basque Country, offers a powerful empirical strategy for evaluating policy interventions when randomized experiments are not feasible. Their pioneering study used a weighted combination of control units to approximate a counterfactual that can be used to assess the impact of treatments or shocks.

Building on this, Abadie et al. (2010) applied the same method to California’s tobacco control, and Abadie et al. (2015) formalized the method as a tool for causal inference in institutional and political economy research.³ SCM constructs a counterfactual using a weighted average of control units that best replicate the treated unit’s pre-treatment trends. This method allows researchers to isolate the effects of a specific reform or event by comparing observed outcomes to an estimate of what would have happened otherwise.

Studies using SCM in high-income countries show that fiscal rules can work to achieve their proposed objectives. For instance, Kraemer and Lehtimäki (2024) apply SCM to assess the European Union’s Stability and Growth Pact, finding that the pact successfully curbed debt accumulation in member countries. Switzerland’s debt brake has also been analyzed (Pfeil and Feld 2024; Salvi et al. 2020), showing that the rule helped lower public debt as a portion of GDP relative to its synthetic counterfactual and improve the cyclically adjusted budget balances. Langer et al. (2025) find that the German 2009 constitutional debt brake reduced budget deficits and did not negatively impact public investment at the federal level. Finally, Herrero-Alcalde et al. (2024) find that the Spanish expenditure rule from 2012 also had a positive effect, improving budget sustainability by limiting public expenditures.

The evidence above demonstrates that fiscal rules can work in countries with strong institutions and robust enforcement mechanisms. Whether these results extrapolate to devel-

³Since its development, SCM has been applied to a wide range of political economy questions. Researchers have used it to study regime changes and leadership shocks (Absher et al. 2020; Cachanosky et al. 2025; Grier and Maynard 2016) and major historical events like the Cuban Revolution (Bastos et al. 2025; Geloso and Pavlik 2021). Beyond regime dynamics, SCM has been applied to analyze constitutional design (Pavlik et al. 2023), EU accession (Miozzi 2024), Quebec’s Quiet Revolution (Geloso and Reilly 2025), the institutional consequences of mass immigration (Powell et al. 2017), and even judicial manipulation (Callais and Mkrtchian 2024). Collectively, this body of work demonstrates SCM’s flexibility as a quasi-experimental tool for evaluating institutional and political shocks that are plausibly exogenous to the status quo.

oping and middle-income countries, where institutional quality may be weaker, is less clear. Another strand of the literature has therefore focused on studying emerging and developing economies and whether fiscal rules bind in these contexts, also using quasi-experimental methods. This research explores whether fiscal rules are binding in middle- and lower-income contexts.

Starting in Africa, Strong (2023) shows that CFA zone debt rules reduced debt ratios, especially in countries with stronger institutions. Moving on to Latin America, Barreix et al. (2019) analyze Panama, Peru, and Colombia adopting rules in 2006, 2003, and 2011 respectively, finding that fiscal rules lower debt-to-GDP ratios in the first two, but that the ratio increased in Colombia following the reform. Arbeláez et al. (2021) also explore Colombia’s 2011 fiscal rule, finding that the adoption of the fiscal rule helped debt remain lower than its counterfactual, opposite to the results found by Barreix et al. (2019). Mendoza Bellido et al. (2021) look at Peru’s fiscal rule using a different treatment year (2000) and find that public investment was reduced after adopting it, with no analysis done on debt outcomes.

The global evidence tends to suggest that well-designed and credible fiscal rules generally improve fiscal outcomes (with some exceptions), especially related to levels of debt. Most importantly, this evidence shows that fiscal rules can still bind even under weaker institutions.

This paper adds to that literature by providing another example of a small, upper-middle-income country with historically weak institutions: Guatemala. It contributes by applying SCM to evaluate how a change in debt-monetization policy, the 1991 Monetary Board resolution and the 1993 constitutional reform, affected debt dynamics.

4 Empirical Strategy

The method chosen to answer the research question is the Synthetic Control Method (SCM). To draw valid causal inferences using the SCM, core assumptions must be satisfied. First is the Stable Unit Treatment Value Assumption (SUTVA), which requires that the treatment applied to the unit of interest does not affect any of the countries in the donor pool. If there are spillover effects, the synthetic counterfactual could be contaminated, biasing the results. This concern is minimal in this context: Guatemala is a relatively small economy with limited regional influence and few financial linkages. A domestic rule on central-bank financing has no clear channel through which it would affect neighboring countries’ debt dynamics.

The second key assumption is the absence of confounding interventions or shocks in the post-treatment period that could also affect the outcome variable. For this study, no

major fiscal reforms occurred in Guatemala that would interfere with the estimation of treatment effects. Two potentially relevant developments, a major central bank reform and the legalization of foreign currency contracts (“Libre Negociación de Divisas”), occurred as late as 2002. However, both changes dealt with monetary policy discretion rather than any fiscal matter, and they do not appear to undermine the fiscal constraints established in the 1990s constitutional reform. They are also far enough from treatment (at least 10 years from the policy change) so that the effects of these policies do not overlap. These identification conditions support the use of SCM as an appropriate framework to estimate the causal effects of Guatemala’s fiscal rules on public debt levels.

While both the 1991 monetary board policy and the 1993 constitutional reform constrained central bank financing of government debt, the 1991 shift represents the first binding policy break. It immediately affected the central government’s ability to accumulate new debt with the central bank, as seen in Figure 1 with the change in trajectory. The 1993 reform created and institutionalized constraints at a higher legal hierarchy but did not produce an additional shift in outstanding debt with the central bank (it kept its downward trajectory). I therefore use 1991 as the treatment for Guatemala’s debt-to-GDP ratio.

5 Data

The dataset spans from 1970 to 2000, providing 21 pre-treatment years (1970 to 1990) and 10 post-treatment years (1991 to 2000) to evaluate the 1991 fiscal rule.⁴ The dependent variable is gross central government debt as a percentage of GDP, drawn from the International Monetary Fund’s (IMF) Global Debt Database (International Monetary Fund 2024). It is defined as the “Total stock of debt liabilities issued by the central government as a share of GDP.” The indicator captures the stock of central government liabilities without netting out financial assets. This indicator normalizes debt levels by economic output, making them comparable across countries and over time. It serves as a meaningful measure of fiscal sustainability.

Countries in the donor pool were selected based on (1) data availability, (2) their similarity to Guatemala in terms of macroeconomic, geographic, and demographic characteristics during the pre-treatment period, and (3) the absence of major structural breaks, institutional crises, or direct exposure to Guatemalan fiscal policy. The donor pool encompasses many Latin American countries, and other developing countries in Africa and Asia. Ultimately, the predictor balance tables presented later in the paper confirm that donor countries closely match Guatemala’s pre-treatment characteristics.

⁴Most empirical applications of SCM use a sufficiently long pre-treatment period of 8 to 15 years.

To improve the precision of the synthetic control, several covariates are used to match on the pre-treatment years, selected based on previous empirical studies on fiscal rules and debt levels (e.g. Arbeláez et al. 2021; Kraemer and Lehtimäki 2024; Salvi et al. 2020). These are the following: real GDP per capita, population, gross fixed capital formation as a percentage of GDP, government spending as a percentage of GDP, inflation, and three lags of the outcome variable (its 1990, 1985, and 1980 values). GDP per capita measures economic development; population captures demographic characteristics; gross fixed capital formation measures investment intensity, which may affect long-term growth and debt sustainability; government spending reflects the size (and scope) of the public sector, which is a key determinant of fiscal behavior; and inflation holds a close relationship with monetary and fiscal policy. All covariates were taken from the World Bank (WB) World Development Indicators (WDI) (World Bank 2024) and Penn World Tables (PWT) (Feenstra et al. 2015). The descriptive statistics for the outcome variable and predictors can be seen in Table 1. For the outcome variable, a small number of missing observations were interpolated linearly to achieve a balanced panel over the pre-treatment period (which is why debt has fewer observations than the other variables in Table 1).

Table 1: Descriptive Statistics: Pre-Treatment Period (1970 to 1990)

Variable	Mean	SD	Min	Max	N
Central Government Debt (% GDP)	35.28	21.07	6.48	112.07	363
GDP per capita	3.40	1.94	0.83	9.33	378
Population (millions)	18.35	24.88	0.12	107.65	378
Gross Fixed Capital Formation (% GDP)	20.33	7.09	4.94	46.22	348
Government Consumption (% GDP)	19.98	7.12	3.34	38.88	378
Inflation Rate (%)	11.90	19.66	-17.63	189.98	365

Notes: Summary statistics for variables used in the synthetic control analysis. Sample includes Guatemala (treated unit, fiscal rule implemented in 1991) and 17 donor pool countries without fiscal rule interventions. Pre-treatment period covers 1970 to 1990 (21 years). GDP per capita in thousands of constant 2017 int'l \$. Population in millions. All other variables in percentages.

6 Results

To assess the impact of Guatemala's reforms, I implement the Synthetic Control Method (SCM) using 1991 as the treatment year. This corresponds to the Monetary Board resolution that limited net government borrowing from the central bank.

Table 2 shows the donor pool weights. Synthetic Guatemala is composed mainly of Belize (28.6%), El Salvador (23.0%), Bangladesh (21.8%), and Paraguay (21.2%), with a

minor contribution from Uganda (5.4%). These countries replicate Guatemala’s pre-1991 debt path closely. Supporting the use of SCM, these are plausible donors: Belize and El Salvador are neighboring Central American countries with similar stages of institutional and economic development, while Bangladesh, Paraguay, and Uganda share characteristics like high reliance on agricultural exports, low tax capacity, and similar pre-treatment trajectories of debt-to-GDP. Table 3 shows the predictor balance. The debt lags are well matched, particularly for the 1990 and 1985 periods. The highest V-weight (0.421) is assigned to 1980 debt levels, reflecting its importance in replicating Guatemala’s pre-treatment trajectory. Some divergence exists in GDP per capita and population, though these receive relatively low weights in the synthetic control construction.

Country	Weight
Belize	0.286
El Salvador	0.230
Bangladesh	0.218
Paraguay	0.212
Uganda	0.054

Table 2: Synthetic Control Weights (Pre-treatment RMSPE: 3.865). Zero weight countries: Botswana, Cameroon, Costa Rica, Honduras, Kenya, Malaysia, Morocco, Pakistan, Senegal, Sri Lanka, Thailand, Tunisia.

Variable	Treated	Synthetic	V-Weight
GDP per Capita	3669.100	2870.738	0.095
Population	7.349	20.139	0.016
Gross Fixed Capital	14.437	16.555	0.211
Gov. Consumption Share	0.132	0.197	0.003
Inflation (Deflator)	12.059	16.570	0.184
Debt (1990)	47.253	42.994	0.018
Debt (1985)	61.247	56.373	0.053
Debt (1980)	15.382	20.697	0.421

Table 3: Covariate Balance and Predictor Weights

Figure 2 shows the evolution of public debt as a share of GDP for Guatemala and its synthetic control from 1970 to 2000. The synthetic unit reproduces Guatemala’s pre-treatment trajectory closely, with a root mean squared prediction error ($\text{RMSPE} = 3.865$). This indicates a strong pre-treatment fit. Starting in 1991, the trajectories of Guatemala and its

synthetic counterpart diverge: synthetic Guatemala maintains debt levels around 35 percent of GDP, while the real Guatemala shows a sharp and sustained decline, hovering around 20 percent of GDP after 1995. Over the 10-year post-treatment period through 2000, the gap averages nearly 12 percentage points of GDP. Upon visual inspection, this suggests that the Monetary Board resolution helped place public finances on a more fiscally conservative path.

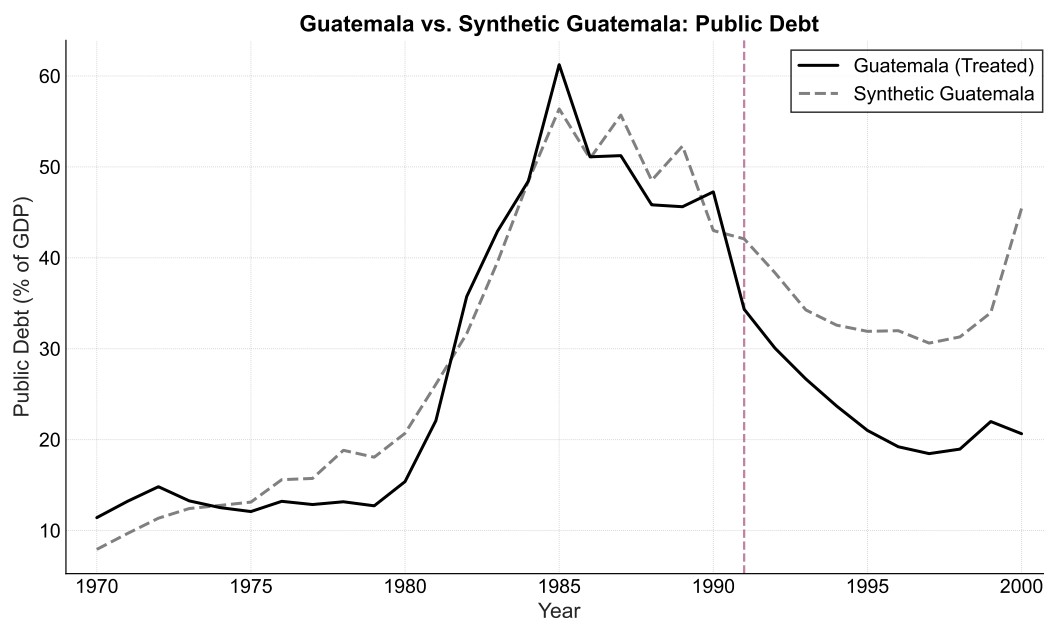


Figure 2: Public Debt as a Percentage of GDP, Using 1991 as Treatment Year

To assess whether the observed gap is due to chance, I examine the statistical significance of the treatment effects. Following Abadie et al. (2010) and Cavallo et al. (2013), inference can be done via permutation tests, where the treatment is reassigned to each control country in the donor pool. This generates a distribution of placebo gaps to compare Guatemala's estimated effect to. Comparing these provides a p-value, which indicates the probability that the effect would arise by mere chance. Figure 3 shows the treatment effects with their corresponding p-values over time. The results indicate that most years in the post-treatment period (1991 to 1999) do not achieve statistical significance, with only the year 2000 showing significance at the 10% level. This means that, even though the effects are negative, there is no strong evidence that the 1991 fiscal rule was effective in reducing debt levels.

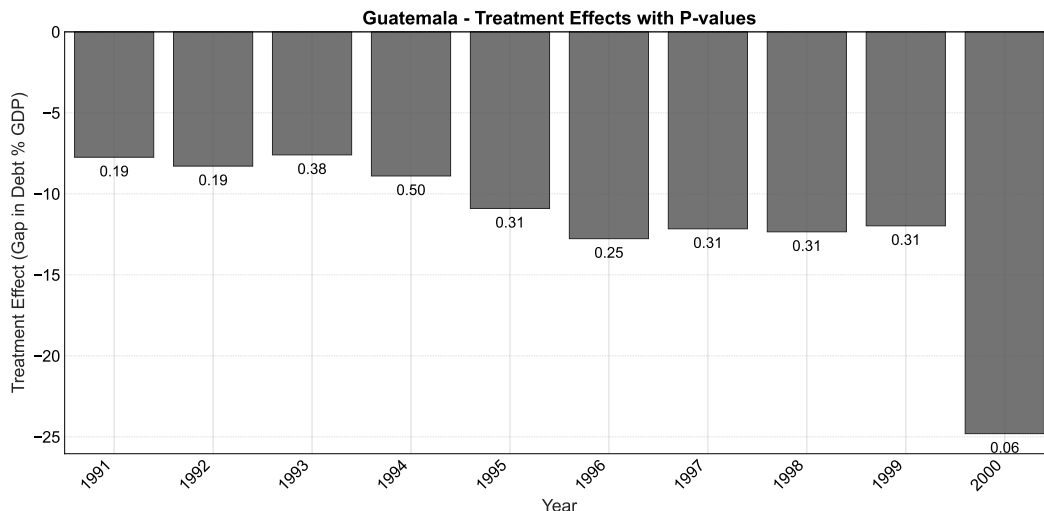


Figure 3: Treatment effects with P-values.

The evidence suggests that the 1991 resolution may have contributed to a shift in Guatemala’s fiscal management. Although the reform was administrative (and it was later taken to a higher level, the constitution), it limited the central government’s ability to accumulate debt through central bank borrowing (debt monetization). However, the lack of statistical significance in most post-treatment years indicates that the observed decline in debt-to-GDP ratios cannot be confidently attributed to the reform alone. An analysis of the underlying debt dynamics is necessary to understand the mechanisms at play, which is the focus of the next section.

As robustness checks for these results, I use two more estimators, reported in the appendices: first, the synthetic difference-in-differences from Arkhangelsky et al. (2021) (Appendix 1); and second, a leave-one-out (jackknife) analysis that drops each donor in turn (Appendix 2). Inference in both is placebo-based, and the results from the main specification hold. There is a gap after treatment, but it is generally not significant.

7 Debt, Inflation, and the Exchange Rate

While the synthetic control analysis shows a decline in debt-to-GDP ratios following the 1991 reform, a closer examination of Guatemala’s debt dynamics reveals a more nuanced story. The synthetic control evaluates IMF gross central government debt as a share of GDP, which for Guatemala runs from roughly 47 percent in 1990 to about 20 percent by the late 1990s. The fiscal rule did successfully reduce central bank holdings of government debt, as intended (shown in Figure 1). However, the observed improvement in debt ratios was

driven primarily by changes in the denominator (nominal GDP growth fueled by inflation) rather than absolute reductions in debt levels.

Figure 4 presents Guatemala’s key macroeconomic series from 1990 to 2000. Panel A indexes the total debt stock, nominal GDP, the price level, and the exchange rate to 1990. Nominal debt continued rising over the decade, but nominal GDP and the price level grew substantially more. Naturally, the debt-to-GDP ratio reduction I observe in the IMF data happens through the denominator rather than through reduced borrowing. Borrowing did shift in composition: the central government’s debt with the central bank declined after 1991 (Figure 1), while its other borrowing grew.

The key to understanding the decline in debt-to-GDP ratios lies in nominal GDP dynamics. Because the ratio is unit-free, its decline reflects nominal GDP growing faster than the nominal stock of debt. In local-currency (quetzal) terms, inflation pushed nominal GDP up faster than debt accumulated, mechanically lowering the ratio; for the foreign-currency share of the debt, the real appreciation of the quetzal further reduced its burden relative to domestic output. Measured in US dollars, this expansion took nominal GDP from approximately \$7.6 billion to \$19.3 billion (World Bank 2024) between 1990 and 2000, an increase of roughly 150 percent. This remarkable expansion was driven largely by inflation rather than real economic growth. The price-level line in Panel A shows that inflation remained elevated throughout the 1990s, averaging approximately 11 percent annually during the post-reform period (1991 to 2000).

What makes Guatemala’s experience particularly interesting is how this inflation occurred alongside relative exchange rate stability. Panel A shows the exchange rate rising from 4.49 GTQ/USD in 1990 to 7.76 GTQ/USD in 2000, a 73 percent cumulative depreciation substantially smaller than cumulative inflation over the same period. The gradual, steady path of the exchange-rate line is consistent with a managed depreciation through what appears to have been a crawling peg regime.

This combination of high domestic inflation with moderate exchange rate depreciation produced a real appreciation of the quetzal. The mechanism through which debt was “inflated away” worked as follows: domestic inflation increased nominal GDP in local currency terms, while the exchange rate depreciated at a slower pace, boosting dollar-denominated GDP. This allowed the debt-to-GDP ratio to fall even as absolute debt levels rose. The central bank’s commitment to exchange rate management, dedicating resources to maintaining a crawling peg rather than allowing free float, facilitated this process.

A key factor enabling this strategy was the substantial growth in remittances. Panel B shows that remittances increased dramatically during this period, both in absolute terms (from \$119 million in 1990 to \$596 million in 2000) and as a share of GDP (rising to 3.1 per-

cent by 2000). This influx of foreign exchange provided critical support for maintaining relative exchange rate stability despite high domestic inflation. The steady flow of remittances helped the central bank sustain its crawling peg regime by augmenting foreign exchange reserves, allowing Guatemala to run higher inflation while avoiding a collapse of the exchange rate. Without this external financing source, the combination of persistent inflation and exchange rate stability observed during the 1990s would likely have been unsustainable.

In summary, the 1991 fiscal rule successfully limited central bank financing of government debt, but the broader improvement in debt-to-GDP ratios stemmed from macroeconomic dynamics rather than fiscal constraint alone. Guatemala managed to reduce its debt burden through inflation while maintaining exchange rate stability, a feat that required central bank commitment to exchange rate management. This reveals an important tension: the institutional reform ended debt monetization in the traditional sense, yet the central bank's role in supporting a crawling peg regime facilitated the inflation process that ultimately reduced the real value of debt.

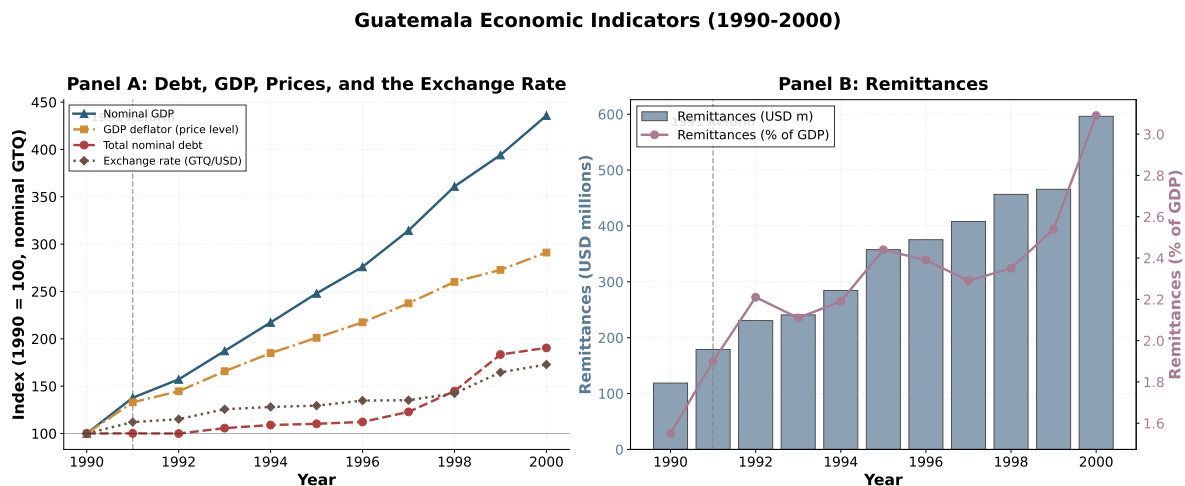


Figure 4: Guatemala Economic Indicators (1990 to 2000)

Notes: Panel A indexes four nominal series to 1990 = 100, all in quetzal terms: the total public debt stock (the consolidated series used in the synthetic control), nominal GDP, the GDP deflator (price level), and the exchange rate (quetzales per US dollar). Both nominal GDP and the price level alone rise faster than the debt stock, so the debt-to-GDP ratio falls through the denominator even as nominal debt grows. Panel B shows remittances in US dollar millions (bars, left axis) and as a share of GDP (line, right axis). The vertical dashed line marks the 1991 Monetary Board resolution.

8 Conclusion and Discussion

This paper examines whether institutional constraints can credibly bind fiscal policy in developing economies with historically weak institutions. I evaluate Guatemala’s 1991 Monetary Board resolution and 1993 constitutional amendment, both of which restricted central government borrowing from the central bank. The analysis yields findings that are more complex than a simple assessment of whether fiscal rules “work” or “fail.”

The synthetic control analysis shows a divergence in debt-to-GDP trajectories beginning in 1991, with Guatemala exhibiting lower ratios relative to its counterfactual through 2000. However, statistical significance is absent across most post-treatment years. The 1993 constitutional reform formalized an existing shift rather than initiating a new policy change.

Central-bank financing of the government, the channel the rule actually restricted, declined steadily after 1991 and was effectively eliminated by the mid-1990s (Banco de Guatemala 2024b). Monetization thus ended. Overall borrowing did not, as the central government’s external debt rose over the same period (Banco de Guatemala 2024c), and total public debt kept increasing. The improvement in debt-to-GDP ratios actually came from the denominator: nominal GDP grew far faster than the debt stock, with inflation driving this expansion.

These findings suggest that institutional reforms can successfully constrain specific financing channels, as evidenced by the elimination of central bank credit to the government. However, the broader fiscal outcomes depend as much on macroeconomic policy choices as on the formal rules themselves. Guatemala’s experience reveals an important tension: the fiscal rule ended traditional debt monetization (which was an important financing option at the time), but the macroeconomic strategy achieved debt reduction through inflation. The central bank’s role in supporting the crawling peg indirectly facilitated the inflation process that reduced the real debt burden.

For policymakers and researchers, Guatemala’s case illustrates both possibilities and limitations of fiscal rules in developing country contexts. Rules can credibly alter specific institutional arrangements and financing mechanisms. Whether such changes translate into improved debt sustainability, however, depends on the broader macroeconomic environment and policy choices regarding inflation, exchange rates, and external factors like remittances or investment flows. The interaction between fiscal rules, monetary policy, and exchange rate regimes deserves more systematic attention in future research.

The inflation-based debt reduction strategy observed in Guatemala raises normative questions that extend beyond the scope of this analysis. While the approach improved the country’s debt-to-GDP position, it redistributed resources through the inflation tax. Whether this represents a desirable or sustainable path for other developing economies facing fiscal

pressures remains an open question.

Appendix 1 Synthetic Difference-in-Differences

The main synthetic control forces the synthetic unit to match Guatemala’s pre-treatment debt level as closely as possible. As a robustness check, I employ the synthetic difference-in-differences (SDID) estimator of Arkhangelsky et al. (2021). SDID permits a constant pre-treatment gap between Guatemala and its synthetic counterpart and matches the trend rather than the level, which makes the estimate less sensitive to imperfect pre-treatment fit.

I apply it to the same outcome (central government debt as a share of GDP), the same donor pool, and the same 1970 to 2000 window used in the main analysis. Because Guatemala is the only treated unit, inference relies on placebo variance.

Table 4 reports the estimate. The SDID average treatment effect is -10.62 percentage points of GDP, close to the average gap recovered by the synthetic control. However, the placebo standard error is large (11.96), and the effect is not significant. Figure 5 plots Guatemala against its SDID counterfactual, and Figure 6 presents the dynamic estimates with their placebo confidence intervals. The post-treatment effects are uniformly negative, but the placebo intervals include zero in all periods. These results are consistent with the main analysis: the direction of the effect is robust, but inference remains insignificant.

Table 4: Synthetic Difference-in-Differences Estimate, 1991 Treatment

Outcome	ATT	Std. Err.	p -value	95% Conf. Interval	
				Lower	Upper
Central government debt (% of GDP)	-10.62	11.96	0.375	-34.07	12.82

Note: SDID estimate following Arkhangelsky et al. (2021). The standard error and p -value are from placebo inference (100 replications), the valid choice for inference with a single treated unit. Sample period: 1970–2000.

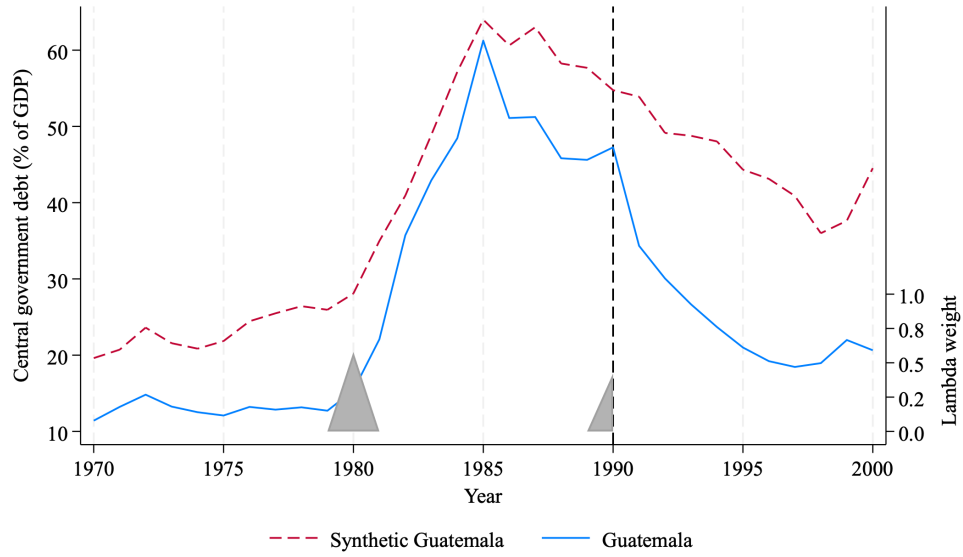


Figure 5: Central Government Debt: Guatemala and its Synthetic Difference-in-Differences Counterfactual.

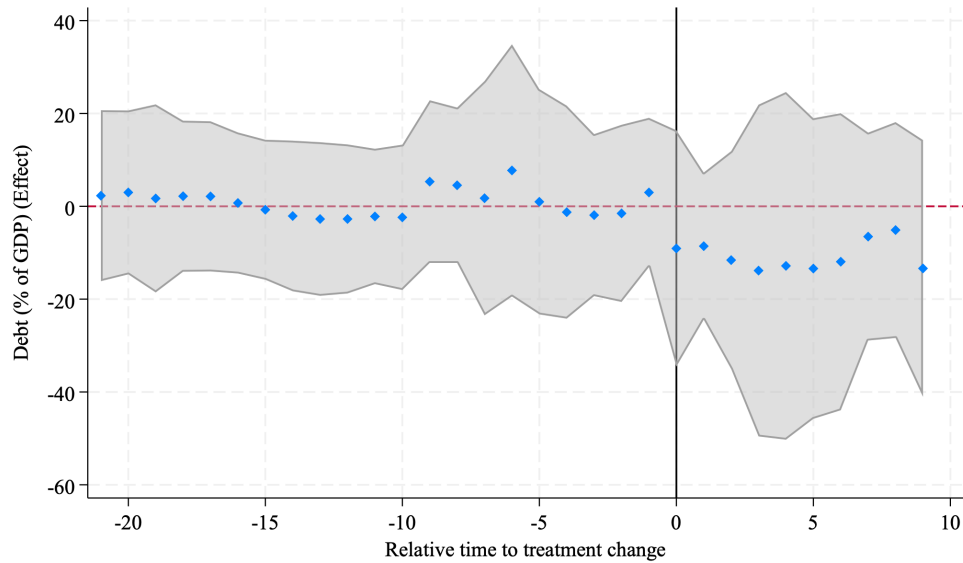


Figure 6: Dynamic SDID Estimates with Placebo Confidence Intervals.

Appendix 2 Leave-One-Out (Jackknife) Robustness

Synthetic Guatemala draws nearly 29 percent of its weight from Belize, which raises the concern that a single donor drives the counterfactual. To evaluate this, I re-estimate the

synthetic control while dropping each donor from the pool one at a time, a leave-one-out (jackknife) procedure. Each re-estimation uses the same outcome, predictors, and optimization as the main analysis. Inference remains placebo-based.

Figure 7 plots the treated series, the full-pool synthetic, and the family of leave-one-out synthetics. The leave-one-out paths cluster tightly around the full-pool counterfactual across the post-treatment period, which indicates that no single donor drives the estimated gap. Figure 8 reports the per-period placebo p -values: the dots mark the full-pool value, the spikes trace the leave-one-out range, and the crosses mark the leave-one-out average. Again, every iteration produces a negative gap. A few iterations reach significance in the early post-treatment years, but on average the leave-one-out estimates are not significant, and no iteration crosses the 10 percent threshold after 1993.

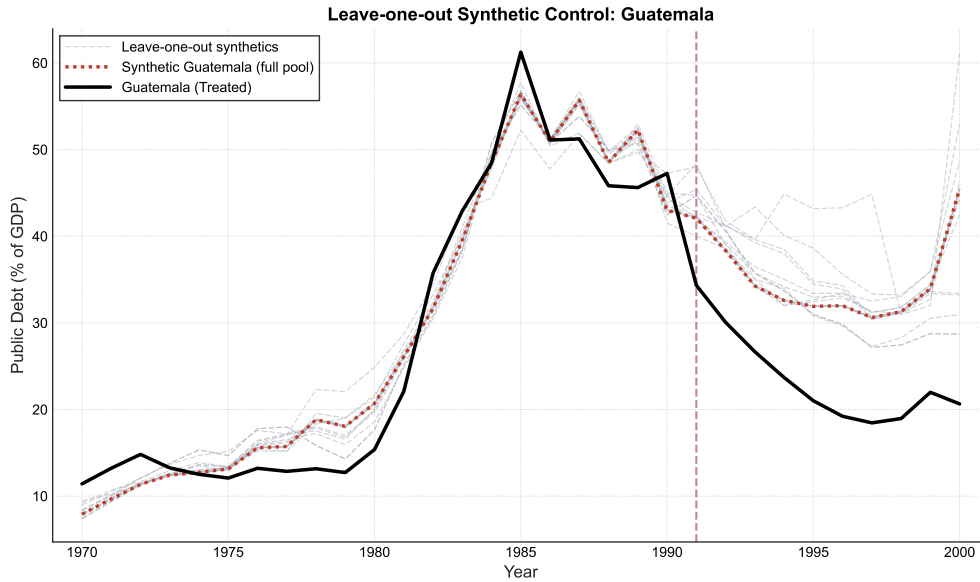


Figure 7: Leave-One-Out Synthetic Control Paths: Guatemala, the Full-Pool Synthetic, and Each Leave-One-Out Synthetic.

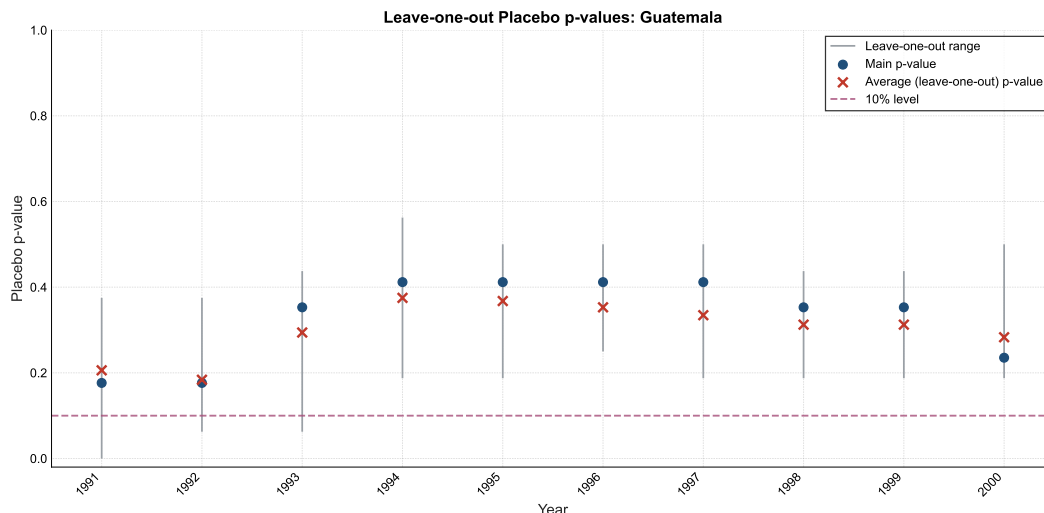


Figure 8: Leave-One-Out Placebo p -values. Dots: full-pool p -value; spikes: leave-one-out range; crosses: average leave-one-out p -value.

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